HOME ASSIGNMENT – 1

Course Code :24CS2202

Name of the Course :Computer Networks.

1. During one transmission, Shraddha wants to transmit the 4-bit data word **1011** using an appropriate **Hamming (7,4) code**.Determine the number of parity bits required and justify your answer.

a. Identify the positions of data bits and parity bits in the Hamming code.
b. Generate the complete 7-bit Hamming code for the given data word using even parity.
c. Assume that during transmission, the received code word is **1010011**.
   - Detect whether an error has occurred.
   - If an error is present, identify the position of the erroneous bit.
d. Correct the error and retrieve the original 4-bit data word.

2. A data communication system uses a **checksum technique** for error detection. The data to be transmitted is divided into **4 blocks**, each consisting of **8 bits**, as shown below:

Block 1: **11011110** Block 2: **10110101** Block 3: **01101100** Block 4: **10011001** Compute the Checksum.

3. The **data bits** to be transmitted are **111011**, and the **generator polynomial** used is

G(x)=x3+1.Explain how the receiver uses the same generator polynomial to detect errors in the received frame..

4. **A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000;**

Show the bit sequence transmitted (in binary) for the four-character frame

"A FLAG B ESC" when each of the following framing methods is used:

(a) **Byte count.**
(b) **Flag bytes with byte stuffing.**
(c) **Starting and ending flag bytes with bit stuffing.**

5. Compare the **OSI reference model** and the **TCP/IP model** in terms of:

- Number of layers
- Functionality of each layer
- Flexibility and standardization

6. Draw a **time–sequence diagram** showing the operation of **Stop-and-Wait ARQ** for transmitting **three frames (F0, F1, F2)** from sender to receiver.

a. Indicate the transmission of frames and acknowledgements.

b. Illustrate the Stop-and-Wait protocol when **frame F2 is corrupted** during transmission.

c. Show retransmission of the frame.

d. Indicate timeout and acknowledgement behavior.

e. Explain why sequence numbers are necessary.